Introduction to Computer and Computer Science

Homework Assignment #3

Due: 10/7 0:00 am

In this homework assignment, you will practice how to create your own functions by solving one of the most difficult issues in your entire life – managing your money for something. As you have learned all the related knowledge during the lectures (if, while, for, def, and so on…), you should have the ability to accomplish this.

Problem #1: Saving your money for the future

Assuming you have graduated from NCTU and now have a great job, your parents start to push you to buy a f***ing house for yourself. Since the current price of house in Taiwan is pretty irrational and ridiculous, you need a plan to save your pay. In this homework, we are going to determine how long it will take to save enough money for paying the down payment.

Before you start to write the code, here are some basic assumptions for this problem:

1. Call the cost of your target **target_price**.
2. Assuming the down payment is 25% of your target price, which leads you this result **down_payment** $= 0.25 \times$ **target_price**.
3. Assuming your monthly salary is **monthly_salary**.
4. Assuming you are going to dedicate a certain percentage of your monthly salary for the down payment. Call it **saving_rate**. Be careful, this number should be in decimal form, i.e. **saving_rate** $= 0.87$ for 87%.
5. Call the amount that you have saved **current_savings**. You should start with **curret_saving**s $= 0$ unless you have wonderful parents or you have "Fa Dà Cái".
6. Assuming you also invest your current savings wisely, with an annual return of **r**. In other words, your account of current savings will receive an additional money **current_savings** $\times$ **(r/12)** at the end of each month. For simplicity, let assume **r** $= 0.04$.

Please write function called "**planA**" that aims to calculate how many months it will take you to save up enough money for the down payment. The parameters of this function **planA** should be:

planA(**target_price**, **monthly_salary**, **saving_rate**)

Please test your function **planA** by asking users to enter this information:

1. Her/his monthly salary
2. The saving rate she/he would like to dedicate
3. The price of her/his target

You can test your function with these cases:

Case 1

Please enter your monthly salary: 60000

How much saving rate would you like to choose: 0.2

How much is your target price: 1000000

Number of months for down payment: 21

Case 2

Please enter your monthly salary: 100000

How much saving rate would you like to choose: 0.38

How much is your target price: 94870000

Number of months for down payment: 339

Please be aware, the answer of the number of months will be slightly different in some cases. An approximate answer (i.e., a close answer) is good enough, you don't need to write a program that prints out a number with exactly same as mine.

Problem #2: Hey, boss! I want more!

In the first problem, we have realized that there is no chance for us to buy a luxury apartment in our entire life, even though you earn 100k per month. However, as an optimistic person, having a wonderful and unreal dream might keep us staying hunger and foolish. Therefore, it is our duty to fight for a better future for our family or those we cared for. As an employee, the easiest way to achieve this objective is asking your boss to raise your salary. Regarding the process of arguing with your boss, let us assume that we have already earned the welfare of raising salary in every six months:

1. Call your semi-annual salary raise **semi_annual_raise**. Again, it should be a decimal number, i.e., **semi_annual_raise** = 0.05 for 5%.
2. Raise your salary by this rate in every six months.

Please write function called "**planB**" that aims to calculate how many months it will take you to save up enough money for the down payment. The parameters of this function **planB** should be:

planB(**target_price**, **monthly_salary**, **saving_rate**, **semi_annual_raise**)

Please test your function **planB** by asking users to enter this information:

1. Her/his monthly salary
2. The saving rate she/he would like to dedicate
3. The price of her/his target
4. The semi-annual salary raise

You can test your function with these cases:

Case 1

Please enter your monthly salary: 60000
How much saving rate would you like to choose: 0.2
How much is your target price: 1000000
How much is your semi-annual salary raise: 0.1
Number of months for down payment: 19

Case 2

Please enter your monthly salary: 100000
How much saving rate would you like to choose: 0.38
How much is your target price: 94870000
How much is your semi-annual salary raise: 0.1
Number of months for down payment: 144

Please be aware, the answer of the number of months will be slightly different in some cases. An approximate answer (i.e., a close answer) is good enough, you don't need to write a program that prints out a number with exactly same as mine.

Hand in procedure:

As we had mentioned in the lecture, you should list all of your collaborators in your programs. Here is the template:

```
# ==========================================================================
#
# Homework assignment #0
# Student: Your name, 你的名字
# Student ID: Your student ID, 你的學號
# Collaborators: John Doe, Jane Doe, ...
#
# ==========================================================================
# Your code goes here...

```

Please save your code as a ".py" file, where the file name should follow this format:

108A_hw#?_ID.py

For example,

108A_hw#1_0180506.py

Please be aware. **We are not going to accept any homework file with wrong file name or without signature**. Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the "New e3" system. There will be a section for uploading your homework.